3.7 F: Exponentials and logarithms

	Content
F1	Know and use the function a^x and its graph, where a is positive. Know and use the function e^x and its graph.

	Content
F2	Know that the gradient of e^{kx} is equal to ke^{kx} and hence understand why the exponential model is suitable in many applications.

	Content
F3	Know and use the definition of $\log_a x$ as the inverse of a^x , where a is positive and $x \geq 0$ Know and use the function $\ln x$ and its graph. Know and use $\ln x$ as the inverse function of e^x

	Content
F4	Understand and use the laws of logarithms: $\log_a x + \log_a y \equiv \log_a(xy)$; $\log_a x - \log_a y \equiv \log_a\left(\frac{x}{y}\right)$; $k\log_a x \equiv \log_a x^k$ (including, for example, $k = -1$ and $k = -\frac{1}{2}$)

	Content
F5	Solve equations of the form $a^x = b$

F6	Use logarithmic graphs to estimate parameters in relationships of the form $y = ax^n$ and $y = kb^x$, given data for x and y.

	Content
F7	Understand and use exponential growth and decay; use in modelling (examples may include the use of e in continuous compound interest, radioactive decay, drug concentration decay, exponential growth as a model for population growth); consideration of limitations and refinements of exponential models.